Zenya Bian • zenyabian@ucla.edu • (248) 781-6811 • Los Angeles, CA • https://zenyabian.github.io

Fourth-year undergraduate student at the University of California, Los Angeles pursuing a career in cell and developmental biology research. Committed to making science more accessible and inclusive for the public.

PUBLICATIONS

- Arabpour, A., DiRusso, J. A., Wu, Q. Y., Larsen, M., Hwang, Y. S., Jacobson, E., Pham, T. X. A., Agranonik, N., Sparrow, M., Monteiro, V. L., **Bian, Z. R.**, Pelaez-Restrepo, N., Callejas-Marin, A., Pasque, V., Plath, K., & Clark, A. T. (2025). TFAP2A+ embryonic progenitor cells undergo fate diversification to give rise to human amnion, germline, and mesoderm. *bioRxiv Preprint*. doi:10.1101/2025.06.26.661839

EDUCATION

University of California, Los Angeles **September 2023 - *June 2027***

B.S. in Molecular, Cell and Developmental Biology (GPA: 3.90 / 4.00)

- Minors in Biomedical Research and Comparative Literature
- Specialization in Computing

HONORS & AWARDS

ABRCMS Student Travel Award **October 2027**
Biomedical Research Summer Scholarship **June 2027**
URC-Sciences Summer Scholarship **June 2027**
Barry Goldwater Scholarship **March 2027**

RESEARCH EXPERIENCE

University of California, Los Angeles **September 2024 - *Present***

Undergraduate Researcher in the Lab of Dr. Amander Clark

- Investigating signaling mechanisms driving germ cell development in the early human embryo.

Oregon Health and Science University **June 2024 - August 2024**

Research Intern in the Lab of Drs. Catherine and James Galbraith

- Developed computational pipeline for cell segmentation and tracking using machine learning and Python.

Undergraduate Research Consortium in Functional Genomics **January 2024 - March 2024**

Undergraduate Researcher instructed by Dr. Carrie Spratford

- Studied effects of systemic oncogenic signaling on hematopoiesis in *Drosophila melanogaster.*

ACADEMIC ACTIVITIES

Integrated and Interdisciplinary Undergraduate Research Program (i2URP) **September 2025 – *Present***

Research Scholar

- Presenting weekly journal clubs and quarterly research talks about biomedical research topics.

UCLA Undergraduate Science Journal **January 2024 - *Present***

Editor-in-Chief

- Leading 60+ editors in publishing 10+ research papers for annual journal volumes.
- Responsible for editor training and instruction on adherence to scientific publishing standards.

Bruins-in-Genomics (B.I.G.) Summer Program **June 2026 – August 2026**

Summer Scholar

- Presenting weekly systems biology journal clubs and attending bioinformatics workshops (Unix, NGS, RNA-Seq).

TEACHING & MENTORING EXPERIENCE

UCLA Center for Education Innovation & Learning in the Sciences **March 2025 - *Present***

Head Life Sciences Learning Assistant

- Assisting student instruction for introductory Python-based biological systems modeling and statistics (leading lab sections, office hours, exam reviews, and programming lessons).

- Facilitating pedagogy instruction for first-time learning assistants.
    - LIFESCI 30A (Mathematics for Life Scientists): Fall 2026, Fall 2025
    - LIFESCI 30B (Mathematics for Life Scientists): Winter 2026
    - LIFESCI 40 (Statistics of Biological Systems): Spring 2026, Spring 2025

PRESENTATIONS & POSTERS

2026 Annual Biomedical Research Conference for Minoritized Scientists (ABRCMS) **November 18-21, 2026**

- **Bian, Z. R.**, Larsen, M., Clark A. *Dynamics in TGF-β, FGF, and WNT Signaling Specify Human Germline and Somatic Lineages.*

2026 Biomedical Research Poster Symposium (Poster) **August 27, 2026**

- **Bian, Z. R.**, Larsen, M., Clark A. *Effects of TGF-β/FGF/WNT Signaling on Human Germ Cell Specification.*

2026 Summer Programs for Undergraduate Research Showcase (Presentation) **August 26, 2026**

- **Bian, Z. R.**, Larsen, M., Clark A. *Signaling Dynamics in Specifying Human Germline and Somatic Lineages.*

Bruins-in-Genomics 2026 Research Symposium (Poster) **August 14, 2026**

- **Bian, Z. R.**, Larsen, M., Clark A. *Effects of TGF-β/FGF/WNT Signaling on Human Germ Cell Specification.*

LGBTQ-Scholars 2026 Annual Research Symposium (Presentation) **May 29, 2026**

- **Bian, Z. R.**, Larsen, M., Clark A. *In Vitro Gametogenesis: Human Reproduction of the Future.*

Broad Stem Cell Research Center (BSCRC) End of Year Core Resource Event (Poster). **May 21, 2026**

- **Bian, Z. R.**, Larsen, M., Clark A. *TGF-β and FGF Signaling Cooperate to Specify Human Germ Cells.*

UCLA Undergraduate Research and Creativity Showcase (Presentation) **May 19, 2026**

- **Bian, Z. R.**, Larsen, M., Clark A. *TGF-β and FGF Signaling Are Required in the Human Germ Cell Specification Niche.*

UCLA Molecular, Cell and Developmental Biology 2026 Research Conference (Poster) **January 29, 2026**

- **Bian, Z. R.**, Larsen, M., Clark A. *Early Combinatorial TGF-β1 and FGF2 Are Required for Germ Cell Specification.*

Annual UCLA Undergraduate Interdisciplinary Research Association (UIRA) Poster Day **May 27, 2025**

- **Bian, Z. R.**, Larsen, M., Arabpour A., Clark A. *Primordial Germ Cells Require Early TGFβ-1 and FGF2 Signaling for Specification.*

SKILLS

Laboratory: Stem Cell Culture • Flow Cytometry • qRT-PCR • Immunofluorescence • Confocal Microscopy • Micropatterning • *Drosophila* husbandry and microdissections
Computational: Python • R • C++ • HTML/CSS • ImageJ (Java) • Unity (C#)
Language: Chinese (fluent) • Japanese (intermediate)

EXTRACURRICULARS

Music Production

- Producing and songwriting science-centered music under the artist name “Oogenesis”
- Featured on the News page for the UCLA Center for Reproductive Science, Health and Education

Lab Vlogging

- Posting short-form videos of wet-laboratory work on TikTok (@future_phd)
- Amassed over 170,000 views since beginning in July 2025